

BUTACU IONEL-CATALIN

linkedin.com/in/catalin-butacu • github.com/CatalinButacu • butacu.catalin@yahoo.com • 0753 320 964

SUMMARY

M.Sc. candidate in Artificial Intelligence with 3 years of production software engineering experience. Applies ML research to real systems—RAG pipelines, on-device quantised models, and cloud data layers—while bringing software engineering discipline (clean architecture, systematic debugging, CI/CD) to applied ML teams.

PROFESSIONAL EXPERIENCE

Gemini CAD Systems

Software Engineer

Iași, RO

June 2023 – Present

- Engineered data-driven features across cloud synchronisation, model/view, and UI layers of a production industrial platform.
- Decomposed complex cloud synchronisation requirements into incremental, testable modules, consistently meeting sprint commitments.
- Reduced technical debt through targeted refactoring, improving codebase robustness and long-term maintainability.
- Diagnosed production crashes and state inconsistencies through systematic root cause analysis.
- Participated in the full Scrum lifecycle: sprint planning, estimation, code reviews, sprint demos, and retrospectives.

Software Engineer Intern

August 2022 – May 2023

- Developed data-driven UI components from design specifications and integrated them into existing production views.
- Delivered pixel-accurate UI and extended feature configurability through import/export workflows.
- Applied full Git workflow: branching strategies, commit conventions, merge requests, code reviews, and CI/CD pipelines.

Special Telecommunications Service (STS)

IT Support Staff — Electoral IT Systems

Botoșani, RO

August 2020 – September 2020

- Provided technical support for electoral IT systems (e.g., SIMPV/SICPV).
- Assisted polling-station staff with application usage and hardware operation.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Data Science Professional (1Z0-1110-25)
- Oracle Cloud Infrastructure 2025 Generative AI Professional (1Z0-1127-25)
- Oracle Cloud Infrastructure 2025 AI Vector Search Professional (1Z0-184-25)
- Oracle Analytics Cloud 2025 Certified Professional (1Z0-1041-25)

CORE SPECIALTIES

Machine Learning: PyTorch, scikit-learn, XGBoost, LightGBM, TFLite, Model Optimisation
AI Systems: LLMs, RAG, Vector Search, Agentic Systems, Prompt Engineering, SSMs
Data & MLOps: Python, Pandas, NumPy, Docker, CI/CD, Oracle Cloud AI, Experiment Tracking

EDUCATION

Gheorghe Asachi Technical University of Iași

Master of Science in Artificial Intelligence

Iași, RO

October 2024 - June 2026

Advanced Coursework: Neural Networks, Machine Learning, Natural Language Processing (NLP), Reinforcement Learning, Multi-Agent Systems, Cloud Computing, Big Data Techniques, Edge Computing Applications, Intelligent Cyber-Security

Bachelor of Science in Automatic Control and Applied Informatics

October 2020 - June 2024

Core Curriculum: Advanced Computer Programming, Data Structures & Algorithms, Database Management Systems, Real-Time Programming, Systems Theory, Microprocessor-Based Systems, Embedded Systems Engineering, Control Engineering, Computer Networks, Neural Networks & Fuzzy Logic, Knowledge-Based Systems

PROJECTS

- M.Sc. Thesis: SSM for Motion Generation** 2026
Developing S4 and Mamba-based architectures for temporally consistent text-to-motion generation (KIT-ML dataset). Incorporated physics constraints to eliminate motion jitter and improve long-range dependency modelling.
- Intelligent IDS: SQL Injection Detection** 2026
Built an ML-powered IDS pipeline integrating Snort with a Python Random Forest classifier. Evaluated across inference latency (<5ms/req), false-positive rate, and cold-start overhead.
- HandSense: NPU Gesture Recognition** 2025
End-to-end YOLOv5n deployment for real-time gesture recognition (HaGRID dataset). Optimised for NPU/CPU with TFLite INT8 quantisation, CLAHE preprocessing, and temporal smoothing.
- PharmaRAG: Local LLM Assistant** 2025
Offline RAG system with FAISS vector search and Mistral-7B for medication retrieval. Chain-of-thought prompting for drug interaction analysis and contraindication detection.
- HeuristicOpt: Optimization Framework** 2025
Python framework with Genetic, Evolutionary, and Stochastic solvers for constraint satisfaction. Benchmarked across convergence speed and solution quality in large-scale search spaces.
- WiDS: fMRI Brain-Age Prediction** (8th National) 2025
ML pipeline for neuroimaging age prediction. Engineered spectral features; ensemble of XGBoost/LightGBM ranked 8th nationally in the WiDS datathon.

VOLUNTEER EXPERIENCE

- LOCAL BEST Group Iași** Iași, RO
IT Department Coordinator Sep 2023 – Aug 2024
- Directed IT operations and maintained technical infrastructure across organizational events.
 - Mentored volunteers and oversaw delivery of technical projects.
 - Administered hardware, software licensing, and network resources.
- Event Coordinator — JobShop Career Event Nov 2022 – Apr 2023
- Coordinated logistics and operations for a large student–industry career event.
 - Managed scheduling, partner engagement, and on-site execution.
 - Improved attendee experience through strategic planning and workflow optimization.
- TOPICS Responsible — Prove It 24H Techathon Dec 2021 – Mar 2022
- Defined technical requirements and coordinated hackathon logistics.
 - Supported participants with rules, resources, and real-time assistance.
 - Facilitated communication between organizers, industry mentors, and teams.
- IT Responsible — BEST Training Weekend Jan 2021 – May 2022
- Developed and maintained the event website using HTML, CSS, and JavaScript.
 - Collaborated with design and marketing teams on content strategy, branding, and technical setup.

ADDITIONAL SKILLS

- Software Eng.:** C++17/20, Qt 5/6, Multithreading, REST APIs, Desktop Applications
- Web & Data:** React, TypeScript, Node.js, Firebase, PostgreSQL, MySQL
- Soft Skills:** Research-to-Product Thinking, Clear Communication, Rapid Self-directed Learning